Day 4 — 100 minus Friends

Module: Friends to Ten (Primary) · **Time:** 35 min

Goal

Children compute `100 − N` for single-digit N using "friends to ten" and a bar model. By end of class: `100 − 7 = 93`, instantly.

Materials

- Bar-model strips: long printed strips divided into 10 boxes of 10 each (= 100 squares total)
- Coloured crayons
- Number cards 0–9

The flow

Settle + chant (5 min)

The friends-to-ten chant. A new line: "*And TEN tens make ONE HUNDRED!*"

Bar-model demo (10 min)

Draw a big bar on the board (or use the strip). Mark off 10 sections of 10 each. Total: 100.

Teacher: *"100 ants on a hundred leaves. 7 ants walk off. How many left?"*

Cross off 7 squares from the rightmost group. *"This last group used to be 10. Now it's 3."*

"How many groups of 10 still left? NINE groups. Plus the 3 leftover ants. So 90 + 3 = 93."

> **The trick:** 7's friend is 3 (10 minus 7 = 3). And there are 9 groups of 10 still standing. So `100 − 7 = 93`.

Demo 3 more, with class predicting:

Problem	Friend of N	9 in front	Answer
100 − 4	6	9	**96**
100 − 8	2	9	**92**
100 − 1	9	9	**99**
100 − 9	1	9	**91**

Activity: bar-model worksheet (15 min)

Each child gets a printed worksheet with 6 bar-model problems:

```
100 − 3 = ?
100 − 5 = ?
100 − 6 = ?
100 − 2 = ?
100 − 8 = ?
100 − 7 = ?
```

For each: shade the right number of squares, then read off the answer. Bonus: write the friend pair next to it.

Wrap-up (5 min)

- Quick fire: teacher calls `100 − N`; children chorus the answer.
- Daily chant.

Homework / take-home

- 10 problems: `100 − 4, 100 − 9, 100 − 6, 100 − 2, 100 − 5, 100 − 8, 100 − 3, 100 − 7, 100 − 1, 100 − 0`. Aim for under 30 seconds.

Differentiation

- **Tier up:** Try `100 − 10`, `100 − 15`, `100 − 20`. Bar model still works.
- **Tier down:** Use only N from 1 to 5. Build up.

Teacher’s quick notes

- The “9 in front” framing is the key insight. Make sure every child says it.
- For N = 0 or N = 10, the bar model is a bit off (no shading, or a whole group gone). Acknowledge and move on; those are edge cases.
- Some children will sniff out the pattern: “to do 100 minus any single digit, just write ‘9’ then the friend of that digit.” Praise — that IS the Nikhilam rule for `100 − N`.